เลขที่นั่งสอบ

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 | **Year:** 2017

Subject: 340152 Electrical Measuring Instruments | **Section:** 5-6

Date: 23 May 2018 | **Time:** 10.00-12.00

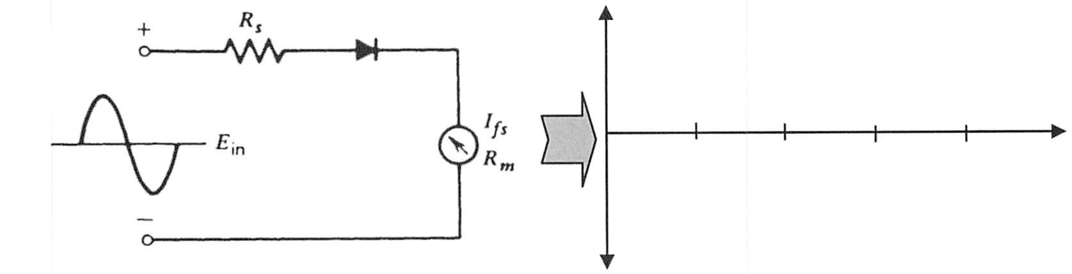
Name________________________________ID: ____________________Class_________

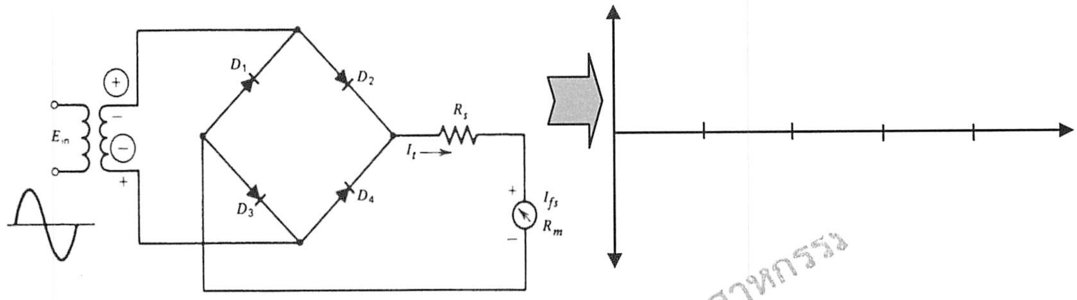
Instructions:

1. The examination has 11 pages (including this page) and a total score of 70 points.
2. Write all your solution and answers on this examination sheet.
3. This is a **closed book** examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor' permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

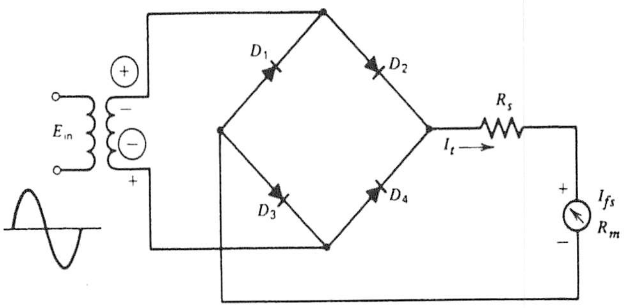
Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Q1) From the circuit, sketch the voltage across the meter movement. (2 marks)





Q2) From the figure, <u>compute</u> the value of the multiplier resistor (R_s) for a 20-Vrms AC range on the voltmeter, where a PMMC instrument with FSD (I_{fs})= 500 μA and meter coil resistance (R_m) is 600 Ω. (3 marks)



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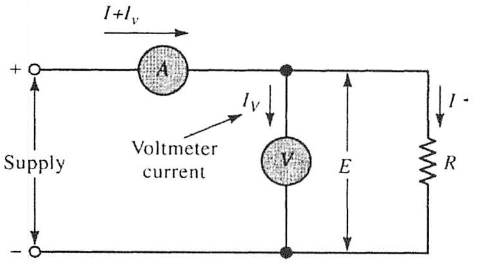
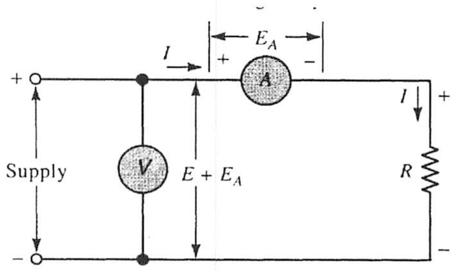
Q3) From the circuit diagrams, <u>identify</u> each of the voltmeter and ammeter methods and <u>explain</u> their operation of both methods of resistance measurement.

(5 marks)





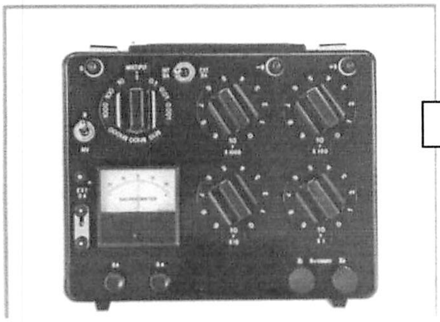
a)

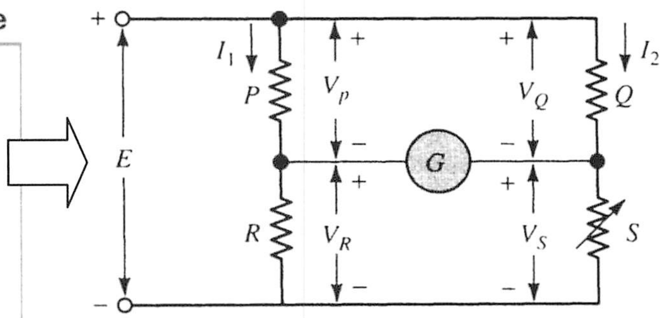
b)

Q4) From the circuit diagram of a wheatstone bridge, explain its operation, and derive its balance equation for R .

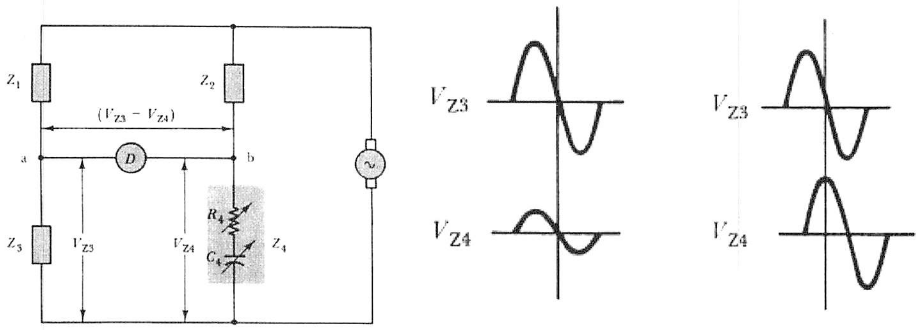
(5 marks)

Portable Wheatstone Bridge



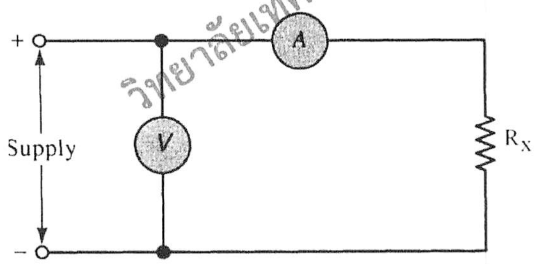


Q5) From the figures, identify each of the case of AC bridge **balanced** or **unbalanced**. (2 marks)



a)................................ b)................................

Q6) A resistance is measured by the ammeter and voltmeter circuit illustrated in the figure. The measured current is 0.5 A, and the voltmeter indication is 500 V. The ammeter has a resistance of R_A = 10 Ω , and the voltmeter on a 1000 V range has a sensitivity of 10 kΩ/V. Calculate the value of R_x by taking the effect of internal resistance (R_A and R_V) into account. (5 marks)



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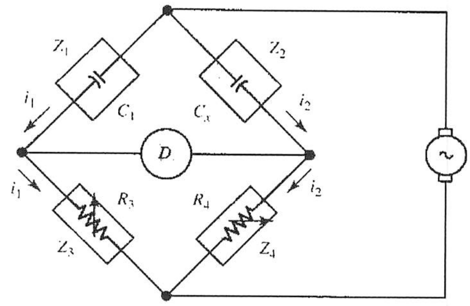
Q7) A simple capacitance bridge, as in Figure, uses a standard capacitor (C_1)= 0.047 μF, and R3/R4 can be set to ratio 50:1 . Calculate the unknown capacitance C_x. (5 points)



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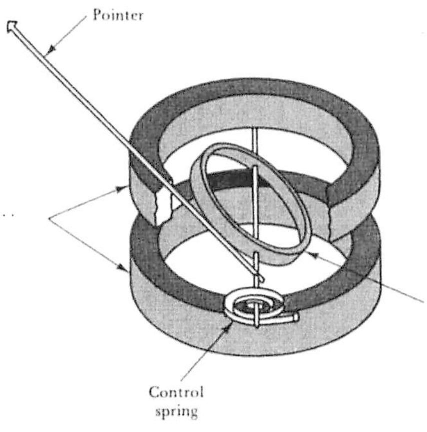
Q8) Identify each part of the electrodynamic instrument and give the major advantage compare with PMMC. (3 marks)

(a)........................



(b)........................

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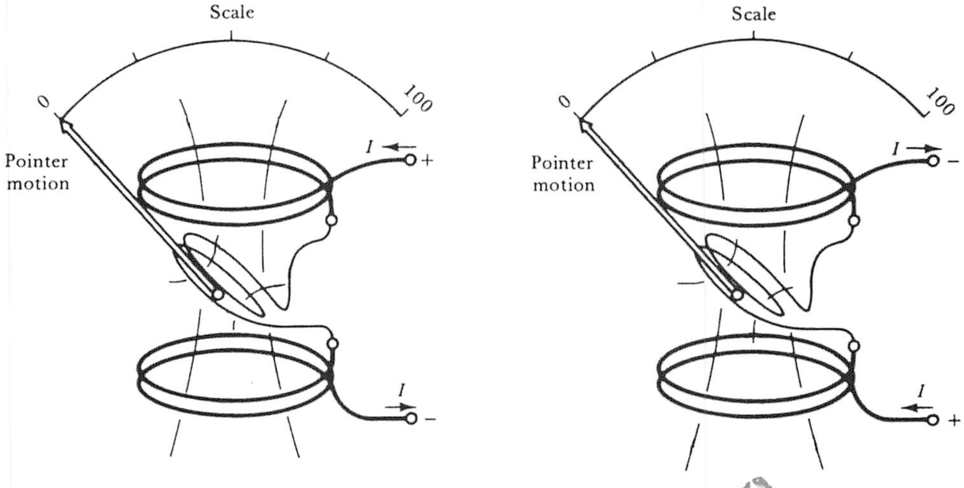
Q9) Describe the basic AC operation of the electrodynamic instrument as shown in the figure. (4marks)



(a) Current flowing from top to bottom

(b) Current flowing from bottom to top

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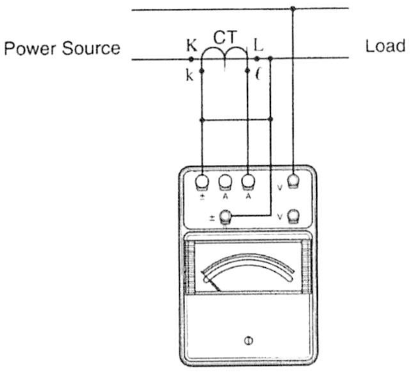
Q10) For the wattmeter, determine the measured power (Watt) where: **(1) Instrument terminal connection = 240 V and 5 A, (2) Indicator Reading = 120, (3) CT ratio = 100/5 A**, respectively. (3 marks)



	Voltage Range	
	120 V	240 V
Current Range	Multiplier Constant	
1 A	1	2
5 A	5	10

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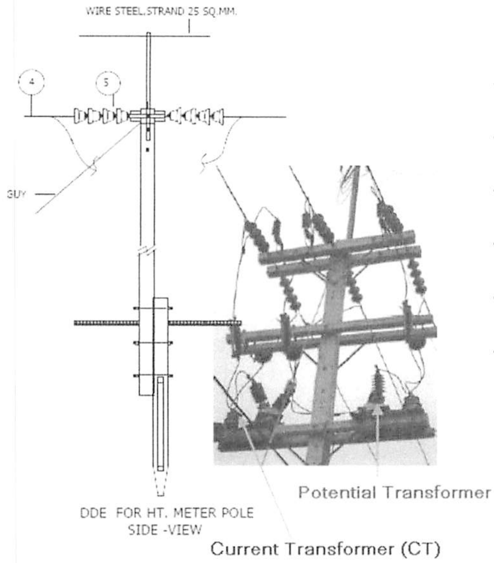
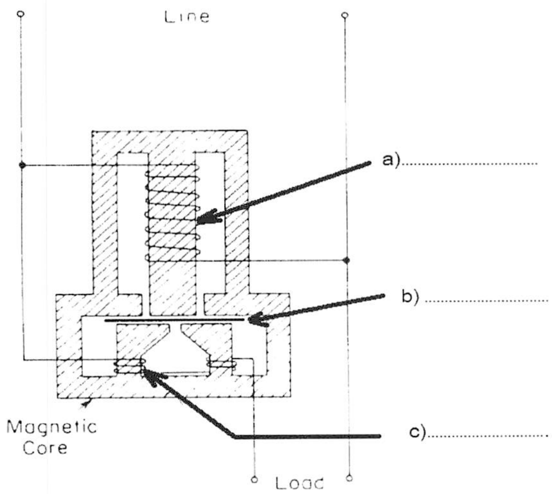
Q11) List main functions of **CT** and **PT**. (3 marks)



Q12) Identify each part of the single phase **watt-hour-meter** and describe its basic concept. (4 marks)



Q13) Draw and explain the simplified block diagram of both ***analog*** and ***digital*** oscilloscopes. (6 marks)

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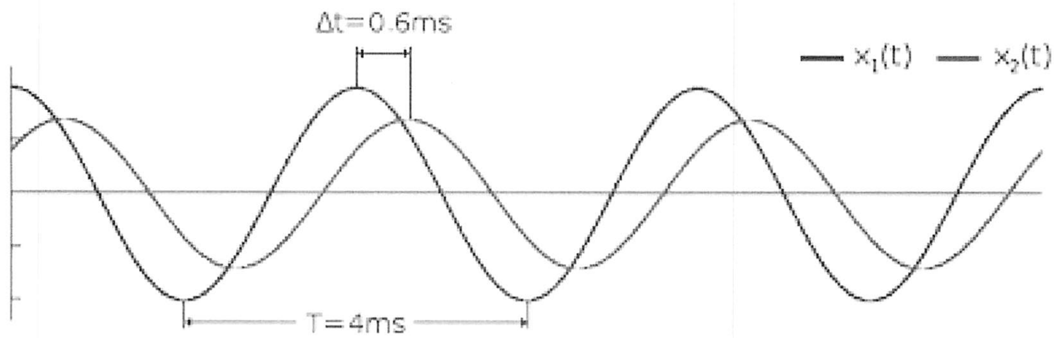
Q14) Determine <u>phase difference</u> between the two waveforms. (3 marks)



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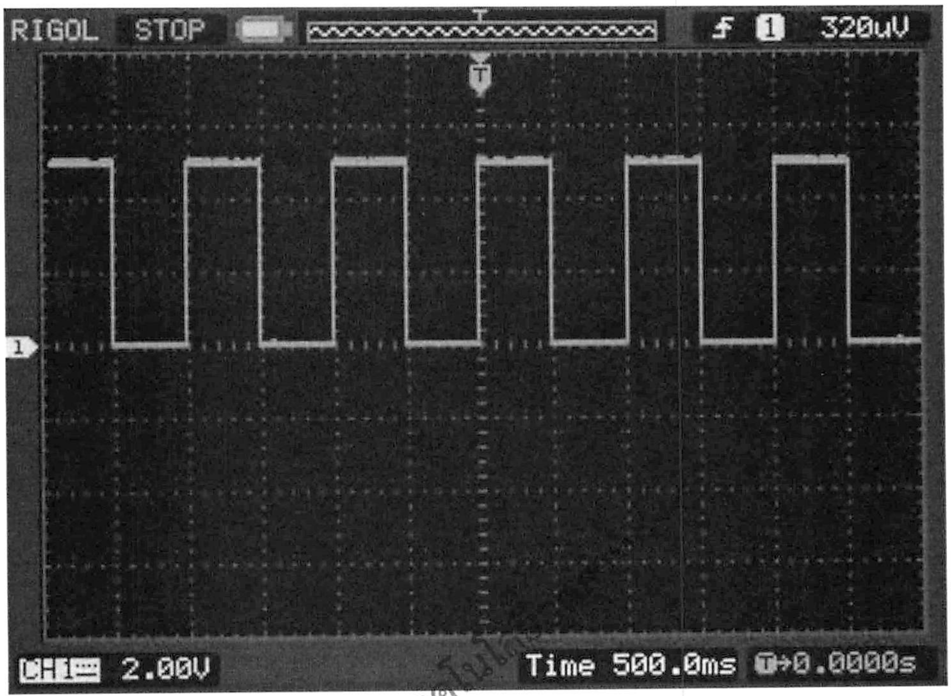
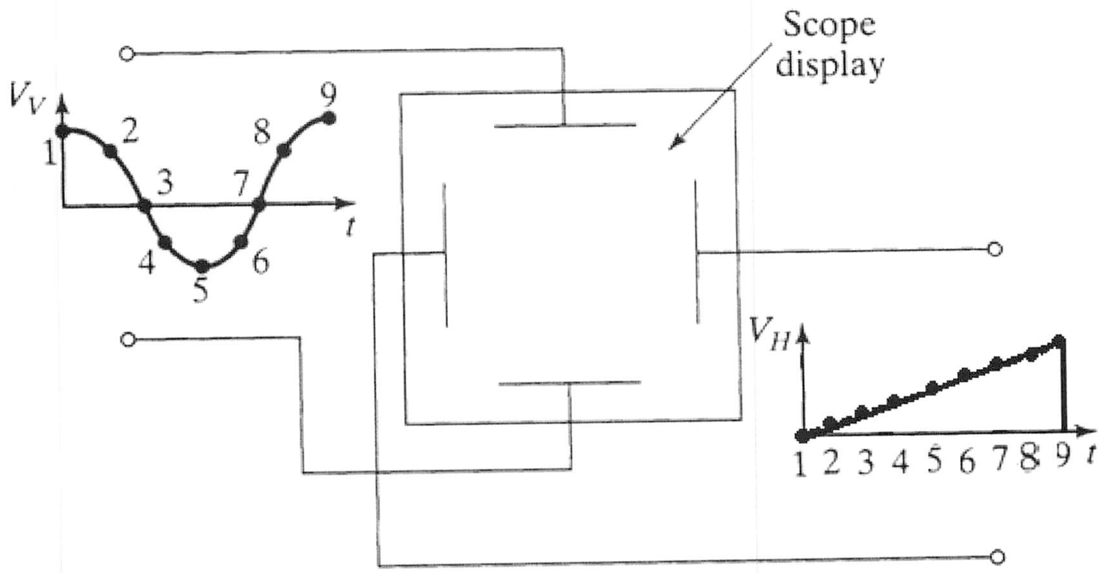
Q15) Determine the following values of the signal waveform where TIME/DIV = 500 ms and VOLTS/DIV = 2 V (3 marks)



a) Pulse amplitude (P_a)........................ b) Pulse Width (PW)....................................

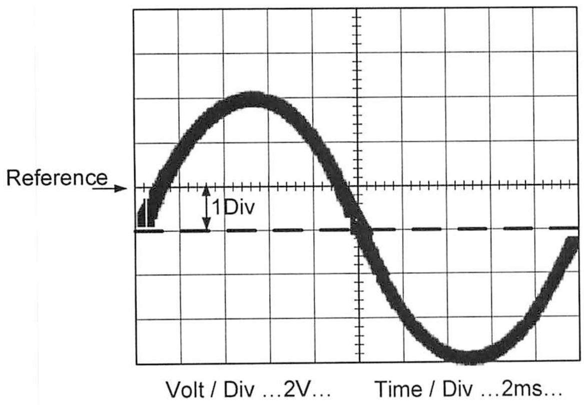
c) Space Width (SW) ....

Q16) Draw the scope display that Lissajous patterns are generated with sine waves to both the horizontal and vertical plates. (4 marks)

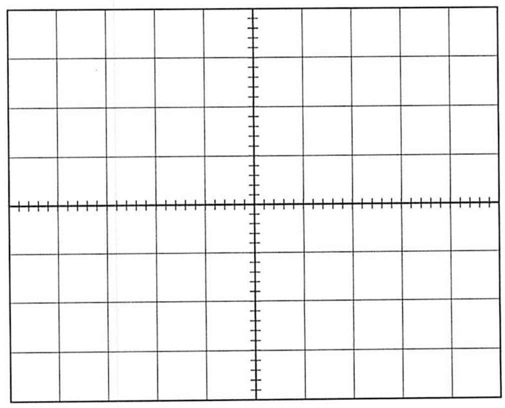


Q17) From the input signal of the oscilloscope, draw the scope display when the setting of **input coupling selector** to AC, DC and **Gnd**. (3 marks)

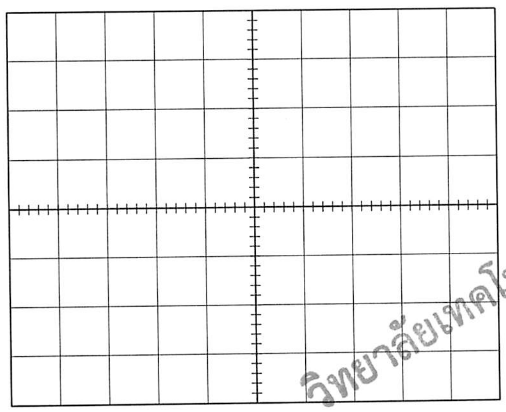
INPUT



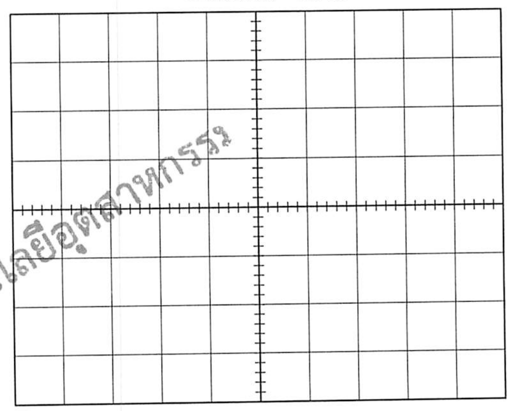
Select: AC



Select: DC



Select: Gnd



Q18) Show how to calibrate <u>DC Instrument</u> such as <u>voltmeter</u> using the standard instruments. (3 marks)

Q19) Draw and explain the **basic function generator** block diagram. (4 marks)

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Assoc. Prof. Dr. Pichet Sriyanyong